Operadic Characterization of Chiral Koszulness: Five Directions and the Shadow Tower Termination Problem

RED-2 (Operadic Homotopy Theory)

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Executive Summary

The monograph's current characterization of chiral Koszulness (Theorem 8.8 = `thm:pbw-koszulness-criterion`) reduces to classical Koszulness of the PBW-associated graded. This is effective—it handles Heisenberg, Kac–Moody, Virasoro, and Yangians—but it is a *detection criterion*, not a *structural explanation*. It says nothing about *why* certain algebras are Koszul, only *that* they are.

The shadow Postnikov tower termination pattern (Heisenberg at arity 2, affine at 3, $\beta\gamma$ at 4, Virasoro at ∞) is the most striking unexplained structure in the monograph. Below I propose five directions for an operadic explanation. My main thesis is:

The termination arity $r_{\max}(\mathcal{A})$ measures the *operadic complexity* of $\mathcal{A}$ as an algebra over the chiral operad $\mathcal{P}^{\mathrm{ch}}$, specifically the arity at which the minimal A_∞-model of $\mathcal{A}$ becomes intrinsically formal.

1 Direction A: Operadic vs. Algebra-Level Koszulness

1.1 What the monograph currently says

The monograph operates at two levels of Koszul duality simultaneously (Remark `rem:operadic-vs-algebraic-` in `algebraic_foundations.tex`, lines 1108–1133):

(a) *Operadic level*: $\mathrm{Com}^! = \mathrm{Lie}$, $\mathrm{Ass}^! = \mathrm{Ass} \otimes \mathrm{sgn}$. The chiral operad $\mathcal{P}^{\mathrm{ch}}$ is identified with $\Omega^*_{\log}(\overline{C}_n(X))$ (Theorem `thm:geometric-bridge`, line 1169), and its Koszul dual cooperad is $(\mathcal{P}^{\mathrm{ch}})^{!\prime}(n) \cong H_*(\overline{C}_n(X))$ (line 1184).

(b) *Algebraic level*: a chiral algebra $\mathcal{A}$ is *chiral Koszul* if the PBW-associated graded $\mathrm{gr}_F\mathcal{A}$ is classically Koszul (Definition `def:chiral-koszul-morphism`, the PBW criterion Theorem `thm:pbw-koszulness-criterion`).

The E_1-chiral Koszul duality theorem (`thm:e1-chiral-koszul-duality`, line 2166) explicitly derives the algebra-level bar-cobar equivalence from the *operadic* Koszulness of Ass via the transfer lemma (`lem:operadic-koszul-transfer`, line 2268). This is the right framework. The question is whether the analogous story works for the full E_∞-chiral (vertex algebra) case.

1.2 Precise conjecture

Conjecture 1 (Operadic inheritance of chiral Koszulness)**.** *Let $\mathcal{P}^{\mathrm{ch}}$ be the chiral operad on a smooth curve X, and let $\mathcal{A}$ be a $\mathcal{P}^{\mathrm{ch}}$-algebra (vertex algebra).*

1. *$\mathcal{P}^{\mathrm{ch}}$ is Koszul as an operad internal to $\mathcal{D}$-mod(X) equipped with the chiral tensor product.*
2. *Every $\mathcal{P}^{\mathrm{ch}}$-algebra $\mathcal{A}$ that is PBW-Koszul in the sense of* `thm:pbw-koszulness-criterion` *is automatically Koszul* as an algebra over a Koszul operad*, in the sense that the operadic bar-cobar adjunction $B_{\mathcal{P}^{\mathrm{ch}}} \dashv \Omega_{\mathcal{P}^{\mathrm{ch}}}$ restricts to an equivalence on the full subcategory of PBW-Koszul algebras.*
3. *Conversely, if $\mathcal{P}^{\mathrm{ch}}$ is Koszul and $\mathcal{A}$ is $\mathcal{P}^{\mathrm{ch}}$-free on generators V modulo quadratic-to-PBW relations, then $\mathcal{A}$ is chiral Koszul.*

Obstacles. The chiral operad $\mathcal{P}^{\mathrm{ch}}$ is not quadratic in the classical sense. Its spaces $\mathcal{P}^{\mathrm{ch}}(n) = \Omega^*_{\log}(\overline{C}_n(X))$ are infinite-dimensional and carry mixed Hodge structures. Koszulness of $\mathcal{P}^{\mathrm{ch}}$ would need to be proved in the enriched sense (internal to $\mathcal{D}$-mod(X)), following the transfer strategy already used for Ass in Lemma `lem:operadic-koszul-transfer`. The key subtlety: $\mathcal{P}^{\mathrm{ch}}$ is *not* the base-change of a classical operad. It is intrinsically geometric. One needs the formality of the FM compactification (Kontsevich) to reduce to the combinatorial operad $H^*(\overline{C}_n)$.

Literature. Francis–Gaitsgory [**?**]: $\mathrm{Lie}^{\mathrm{ch}} \dashv \mathrm{Com}^{\mathrm{ch}}$ adjunction on $\mathrm{Ran}(X)$. Loday–Vallette [**?**], Theorem 11.4.1: bar-cobar for algebras over Koszul operads. Fresse [**?**], §16: model-categorical bar-cobar for general symmetric monoidal model categories. The monograph's Lemma `lem:operadic-koszul-transfer` is the precise bridge.

Utility Assessment. **High utility, moderate difficulty.** This conjecture would unify the PBW criterion with the operadic framework and explain *why* the PBW strategy works: it works because $\mathcal{P}^{\mathrm{ch}}$ is Koszul, and PBW-flat algebras over Koszul operads are automatically Koszul. The proof strategy is essentially already in the monograph for Ass; extending it to $\mathcal{P}^{\mathrm{ch}}$ requires the formality input.

2 Direction B: Distributive Law Characterization

2.1 The idea

A vertex algebra OPE has the structure

$$a(z)b(w) = \underbrace{\sum_{n\geq 0} \frac{(a_{(n)}b)(w)}{(z-w)^{n+1}}}_{\text{singular part}} + \underbrace{:a(z)b(w):}_{\text{regular part}}.$$

The singular part encodes the "Lie-type" data (poles, commutators), and the regular part encodes the "commutative-type" data (normal ordering). The PBW theorem for vertex algebras says that the multiplication map

$$\mathrm{Sym}^{\mathrm{ch}}(V) \xrightarrow{\sim} \mathrm{gr}_F \mathcal{A}$$

is an isomorphism. In the Loday–Vallette framework, PBW theorems for algebras are consequences of *distributive laws* between operads: if $P = P_1 \circ_\lambda P_2$ (a distributive law $\lambda : P_2 \circ P_1 \to P_1 \circ P_2$), then free P-algebras have PBW bases indexed by P_1-monomials applied to P_2-monomials.

2.2 Precise conjecture

Conjecture 2 (Distributive law for chiral Koszulness)**.** *There exist sub-operads $\mathcal{P}^{\mathrm{ch}}_{\mathrm{sing}} \hookrightarrow \mathcal{P}^{\mathrm{ch}}$ and $\mathcal{P}^{\mathrm{ch}}_{\mathrm{reg}} \hookrightarrow \mathcal{P}^{\mathrm{ch}}$ encoding, respectively, the singular (OPE pole) and regular (normal ordering) parts of the chiral product, together with a distributive law*

$$\lambda : \mathcal{P}^{\mathrm{ch}}_{\mathrm{reg}} \circ \mathcal{P}^{\mathrm{ch}}_{\mathrm{sing}} \longrightarrow \mathcal{P}^{\mathrm{ch}}_{\mathrm{sing}} \circ \mathcal{P}^{\mathrm{ch}}_{\mathrm{reg}}$$

such that:

1. *$\mathcal{P}^{\mathrm{ch}} \cong \mathcal{P}^{\mathrm{ch}}_{\mathrm{sing}} \circ_\lambda \mathcal{P}^{\mathrm{ch}}_{\mathrm{reg}}$ as operads.*
2. *The PBW theorem for chiral algebras (the isomorphism $\mathrm{Sym}^{\mathrm{ch}}(V) \cong \mathrm{gr}_F \mathcal{A}$) is a consequence of this distributive law.*
3. *A chiral algebra $\mathcal{A}$ is Koszul if and only if both $\mathcal{P}^{\mathrm{ch}}_{\mathrm{sing}}$ and $\mathcal{P}^{\mathrm{ch}}_{\mathrm{reg}}$ are Koszul and the distributive law is compatible with the Koszul resolutions.*

Obstacles. This is the most speculative direction. The main difficulty is that the singular/regular decomposition of the OPE is *not operadic* in a naive sense: the singular part at n points involves pole orders that depend on the specific algebra $\mathcal{A}$ (e.g., double pole for Heisenberg, quartic pole for Virasoro). A genuine distributive law would need to operate at the level of the chiral operad itself, not at the algebra level.

Moreover, Loday–Vallette's distributive law theory ([**?**], §8.6) applies to quadratic operads and produces quadratic-linear algebras via the rewriting method. The chiral setting involves poles of *all* orders, which goes beyond the quadratic-linear framework.

A more realistic version: the distributive law exists at the level of the *PBW-graded* chiral operad $\mathrm{gr}_F \mathcal{P}^{\mathrm{ch}}$, where it reduces to the classical $\mathrm{Com} \circ \mathrm{Lie}$ distributive law (encoding the PBW theorem for universal enveloping algebras). The failure of the distributive law to lift to the ungraded level is measured by a sequence of obstructions, and the *shadow tower* is precisely this obstruction sequence.

Literature. Loday–Vallette [**?**], §8.6: distributive laws and PBW. Dotsenko–Khoroshkin [**?**]: Gröbner bases for operads, rewriting approach to PBW. Bremner–Dotsenko: algebraic operads and rewriting systems. For the vertex algebra PBW theorem: Li [**?**], Frenkel–Ben-Zvi [**?**].

Utility Assessment. **Medium utility, high difficulty.** If achieved, this would give the *deepest* structural explanation of chiral Koszulness. But the gap between the operadic distributive law framework and the analytic structure of OPE poles is large. The modified version (distributive law on the associated graded, with the shadow tower as the obstruction to lifting) is more realistic and connects directly to the monograph's existing architecture.

3 Direction C: Modular Koszulness and Shadow Tower Termination

3.1 The key data

From the monograph (Example `ex:tower-four-archetypes`, `higher_genus_modular_koszul.tex` lines 8571–8597):

Algebra	κ	$\mathfrak{C}$	$\mathfrak{Q}$	$r_{\max}$
Heisenberg $\mathcal{H}_k$	k	0	0	2
Affine $\widehat{\mathfrak{sl}}_2$	κ	$\kappa(x,[y,z])$	0 (Jacobi)	3
$\beta\gamma$	0	0	$\mathrm{cyc}(m_3)$	4
Virasoro Vir_c	$c/2$	$2x^3$	$\frac{10}{c(5c+22)}x^4$	∞

The shadow Postnikov tower (Definition `def:shadow-postnikov-tower`) is a sequence of MC truncations $\Theta_{\mathcal{A}}^{\leq r}$ in the modular convolution dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$, where $\Theta_{\mathcal{A}}^{\leq r}$ solves MC modulo weight $r+1$. The obstruction class $o_{r+1}(\mathcal{A}) \in H^2(\mathcal{A}_{r+1,0}^{\mathrm{sh}})$ controls extension. Termination at $r_{\max}$ means $\mathcal{A}_{r,0}^{\mathrm{sh}} = 0$ for $r > r_{\max}$.

3.2 Operadic explanation of termination

Conjecture 3 (Operadic complexity controls termination)**.** *The shadow tower termination arity $r_{\max}(\mathcal{A})$ equals the* operadic complexity *of $\mathcal{A}$, defined as the minimal arity r such that the transferred A_∞ operations $m_n^{\mathrm{tr}} = 0$ for all $n > r$ on the minimal model of $\mathcal{A}$ as a $\mathcal{P}^{\mathrm{ch}}$-algebra.*

Here is the proposed dictionary:

Algebra	**Minimal model**	$r_{\max}$
Heisenberg	Strictly commutative ($m_n = 0$, $n \geq 3$)	2
Affine KM	Strict Lie ($m_3 \neq 0$ from bracket, $m_n = 0$, $n \geq 4$)	3
$\beta\gamma$	A_∞ with m_3 and m_4 from contact terms	4
Virasoro	Full A_∞ tower, $m_n \neq 0$ for all n	∞

Why this should be true. The shadow at arity r is built from graph sums (Definition `def:modular-bar-hamiltonian`, line 8610) whose vertices carry the transferred operations $\ell_{|v|}^{\mathrm{tr}}$ from the cyclic minimal model. If the minimal model has $m_n^{\mathrm{tr}} = 0$ for $n > r$, then all graph sums with a vertex of valence $> r$ vanish, and the only contributions to the shadow at arity $> r$ come from graphs whose vertices all have valence $\leq r$. The obstruction class o_{r+1} involves the bracket $[\Theta^{\leq r}, \Theta^{\leq r}]_{r+1}$, which is a sum over graphs with two vertices sewn together. If the vertex operations truncate at arity r, and the bracket produces arity $r+1$ only by combining vertices of arity $\leq r$, then $o_{r+1} = 0$ if and only if the "Jacobi-type" identity for these lower-arity operations holds.

The pattern:

- **Heisenberg**: $m_n^{\mathrm{tr}} = 0$ for $n \geq 3$ (abelian). All graph sums with ≥ 3-valent vertices vanish. The modular content is purely from edge contractions (propagators), giving the Gaussian $\kappa \cdot \lambda_g$ formula.

- **Affine**: m_2^{tr} is the Lie bracket, $m_3^{\mathrm{tr}} = 0$ by transfer from the Koszul resolution (the A_∞ structure on $\mathrm{Ext}^*(\mathbb{C}, \mathbb{C})$ for $U(\mathfrak{g})$ has $m_3 = 0$ because the Koszul resolution is *minimal* and the Chevalley–Eilenberg complex is formal through degree 3). The cubic shadow $\mathfrak{C}$ comes from m_2 composed with itself (the s-t-u channels), and the quartic obstruction vanishes by the Jacobi identity.

- $\beta\gamma$: The normal-ordered product $:\beta\gamma:$ gives a nontrivial m_3^{tr} (the contact term from the quartic singularity in the $\beta\gamma$ OPE after subtracting the simple pole), but m_4^{tr} is exact (killed by rank-one abelian rigidity, `thm:nms-rank-one-rigidity`).

- **Virasoro**: The composite field $:TT:$ and its descendants generate an infinite sequence of nonvanishing transferred operations. The A_∞ minimal model of Vir_c is genuinely infinite: $m_n^{\mathrm{tr}} \neq 0$ for all n because the OPE involves a quartic pole, composite fields of arbitrarily high weight, and the central charge enters nonlinearly at every order (as noted in `chiral_koszul_pairs.tex`, line 2896). The quintic is forced because $\{\mathfrak{C}, \mathfrak{Q}\}_H \neq 0$ (Theorem `thm:nms-virasoro-quintic-forced`).

Obstacles. The precise relationship between m_n^{tr} and the shadow obstruction o_n is not a direct equality. The obstruction o_{r+1} involves *all* graphs of weight $r+1$, including graphs with low-valence vertices connected by many edges (loops). A vertex with $m_k^{\mathrm{tr}} = 0$ for $k > r$ does not immediately imply $o_{r+1} = 0$, because multi-loop graphs with vertices of valence $\leq r$ can contribute to weight $r+1$.

The precise statement would need: if $m_k^{\mathrm{tr}} = 0$ for $k > r$ AND the genus-loop operator Λ_P acting on arity-$(r+2)$ shadows from lower-arity vertices gives zero, THEN $o_{r+1} = 0$. The genus-loop condition is an additional constraint beyond the truncation of the minimal model.

Literature. Merkulov [**?**]: minimal models and A_∞ transfer for operadic algebras. Kadeishvili [**?**]: A_∞ minimal models. Kontsevich–Soibelman [**?**]: formal moduli from L_∞ and A_∞. The graph sum formula (`def:modular-bar-hamiltonian`) is the Feynman expansion of the quantum L_∞ structure (Chuang–Lazarev [**?**]).

Utility Assessment. **Very high utility, moderate difficulty.** This is the most promising direction. It gives a precise operadic meaning to the shadow termination pattern and connects it to the classical theory of A_∞ minimal models. The main work is making the graph-sum argument rigorous: showing that truncation of the minimal model controls termination of the shadow tower, accounting for the genus-loop contributions.

This could be stated as a theorem in the monograph, not merely a conjecture, if the graph-sum analysis is carried out carefully.

4 Direction D: Koszul Duality of the Chiral Operad

4.1 What is $(\mathcal{P}^{\mathrm{ch}})^!$?

The monograph identifies $\mathcal{P}^{\mathrm{ch}}(n) \cong \Omega^*_{\log}(\overline{C}_n(X))$ (`algebraic_foundations.tex`, line 1171) and the Koszul dual cooperad $(\mathcal{P}^{\mathrm{ch}})^{!\prime}(n) \cong H_*(\overline{C}_n(X))$ (line 1184). Since $H^*(\overline{C}_n(\mathbb{C}))$ is generated by the Arnold classes ω_{ij} with the Arnold relations, and the homology $H_*(\overline{C}_n(\mathbb{C}))$ is generated by the dual classes α_{ij} with the dual relations, the Koszul dual operad $(\mathcal{P}^{\mathrm{ch}})^!$ is:

$$(\mathcal{P}^{\mathrm{ch}})^!(n) = H_*(\overline{C}_n(X), \mathbb{C}) = \text{the Lie-type operad generated by "collision classes".}$$

On $\mathbb{P}^1$: $H^*(\overline{C}_n(\mathbb{P}^1)) \cong \mathrm{Com}(n)$ (the commutative operad), and $H_*(\overline{C}_n(\mathbb{P}^1)) \cong \mathrm{Lie}(n)$ (by the partition lattice theorem, `algebraic_foundations.tex`, Theorem `thm:com-lie`, line 1236). So:

$$(\mathcal{P}^{\mathrm{ch}})^!\Big|_{X=\mathbb{P}^1} \cong \mathrm{Lie}^{\mathrm{ch}}.$$

This recovers the Francis–Gaitsgory $\mathrm{Lie}^{\mathrm{ch}} \dashv \mathrm{Com}^{\mathrm{ch}}$ adjunction.

4.2 Precise question

Question 1. *For a higher-genus curve X (or the universal curve over $\overline{\mathcal{M}}_{g,n}$), what is $(\mathcal{P}_X^{\mathrm{ch}})^!$? Does it depend on the complex structure of X? Is the Koszul dual cooperad $(\mathcal{P}_X^{\mathrm{ch}})^{!\prime}(n) = H_*(\overline{C}_n(X))$ formal (i.e., does the cooperad structure on chains $C_*(\overline{C}_n(X))$ transfer to cohomology)?*

At genus $g \geq 1$, $H^1(\overline{C}_n(X)) \neq 0$ (from the homology of the curve itself), and the chiral operad $\mathcal{P}_X^{\mathrm{ch}}$ is *not* the base-change of Com. The Koszul dual $(\mathcal{P}_X^{\mathrm{ch}})^!$ would be a *genus-dependent* Lie-type operad, whose arity-n spaces encode the interaction between the OPE and the topology of X.

Conjecture 4 (Genus-dependent Koszul dual)**.** *For a smooth curve X of genus g:*

1. *The cooperad $\{C_*(\overline{C}_n(X))\}$ is formal if and only if $g = 0$.*
2. *At genus $g \geq 1$, the non-formality is controlled by the period matrix $\Omega_g \in \mathfrak{H}_g$, and the resulting "Lie-type" operad $(\mathcal{P}_X^{\mathrm{ch}})^!$ carries a connection over the Siegel upper half-space (equivalently, over $\overline{\mathcal{M}}_g$).*
3. *The curvature of this connection is the modular characteristic $\kappa(\mathcal{A})$: the obstruction to $(\mathcal{P}_X^{\mathrm{ch}})^!$ being independent of the complex structure.*

This is essentially saying: the genus-g bar complex $\bar{B}^{(g)}(\mathcal{A})$ computes the $(\mathcal{P}_X^{\mathrm{ch}})^!$-coalgebra structure on $\mathcal{A}$, and the curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ is the failure of this coalgebra structure to be flat over $\overline{\mathcal{M}}_g$.

Obstacles. The formality of configuration spaces on higher-genus curves is a deep open question. On $\mathbb{C}$ (or $\mathbb{P}^1$), Kontsevich formality gives $C^*(\overline{C}_n(\mathbb{C})) \simeq H^*(\overline{C}_n(\mathbb{C}))$ as operads. On Σ_g for $g \geq 1$, this fails: $C^*(\mathrm{Conf}_n(\Sigma_g))$ is *not* formal in general (there are nontrivial Massey products from the interaction of the curve's homology with the configuration space structure). This non-formality is closely related to the non-vanishing of the shadow tower at higher arities.

Utility Assessment. **High utility, high difficulty.** This direction connects the shadow tower directly to the homotopy theory of configuration spaces on curves, which is an active area of research. The payoff would be enormous: a geometric explanation of κ as the curvature of a connection on the Koszul dual operad. But the technical machinery (formality on curves, connections on operads over moduli) is demanding.

5 Direction E: Transfer, Formality, and Operadic Koszulness

5.1 The framework

The monograph states that bar-cobar preserves quasi-isomorphisms "because it is a quantum L_∞ functor" (`CLAUDE.md`). More precisely: the convolution L_∞-algebra $\hom_\alpha(C, A)$ of Robert-Nicoud–Wierstra carries a transferred L_∞ structure, and the bar-cobar adjunction is the MC moduli of this structure (Theorem `thm:modular-quantum-linfty`, line 7537).

The homotopy transfer theorem (HTT) says: given a deformation retract $V \hookrightarrow A \twoheadrightarrow V$ with $V = H^*(A)$, the A_∞ structure on A transfers to a *minimal* A_∞ structure on V. The transferred operations m_n^{tr} are sums over planar trees with n leaves, each internal edge decorated by the contracting homotopy h and each vertex by the original operations.

Conjecture 5 (Formality characterization of Koszulness)**.** *A chiral algebra $\mathcal{A}$ is Koszul if and only if the convolution L_∞-algebra* $\hom_\alpha(\{C_*(\overline{C}_n(X))\}, \mathrm{End}_{\mathcal{A}})$ *is* formal*: i.e., quasi-isomorphic to its cohomology (a dg Lie algebra with zero differential) as an L_∞-algebra.*

Moreover:

1. *$\mathcal{A}$ is Koszul with $r_{\max} = 2$ (shadow tower terminates at arity 2) if and only if the convolution L_∞-algebra is* intrinsically formal *(all higher brackets $\ell_k = 0$ for $k \geq 3$ on cohomology).*

2. *$r_{\max} = r$ if and only if the convolution L_∞-algebra is* formal through arity r *but not through arity $r+1$: the transferred brackets ℓ_k^{tr} vanish for $k > r$ but ℓ_{r+1}^{tr} represents a nontrivial class in the Harrison cohomology of the L_∞ deformation complex.*

3. *$r_{\max} = \infty$ (non-termination, e.g., Virasoro) if and only if the convolution L_∞-algebra is* not formal at any finite level*: there exist nontrivial transferred brackets at every arity.*

Observation 1 (The formality obstruction tower IS the shadow tower)**.** *The obstructions to formality of an L_∞-algebra are organized by a Postnikov-type tower: at each stage, one tries to kill the lowest-arity nontrivial transferred bracket by a quasi-isomorphism. The obstruction to killing ℓ_n^{tr} lives in a Harrison-type cohomology group. This tower is formally identical to the shadow Postnikov tower:*

- *$\Theta^{\leq 2} = \kappa \leftrightarrow$ binary bracket (dg Lie structure).*
- *$o_3 =$ cubic shadow obstruction $\leftrightarrow \ell_3^{\mathrm{tr}}$, the ternary Massey-type product.*
- *$o_4 =$ quartic resonance $\leftrightarrow \ell_4^{\mathrm{tr}}$, the quaternary obstruction.*

This is consistent with the monograph's explicit formula for $\ell_3^{(0)}$ (Construction `constr:explicit-convoluti` line 7582): it is the sum over the three binary trees with 3 leaves (s, t, u channels), each internal edge contracted with the homotopy propagator. On cohomology, $[\ell_3^{(0)}] = 0$ in $H^*(\overline{\mathcal{M}}_{0,4}) = H^*(\mathbb{P}^1)$ because $H^1(\mathbb{P}^1) = 0$. But at the chain level, $\ell_3^{(0)}$ is nontrivial and feeds the cubic shadow.

The quaternary bracket $\ell_4^{(0)}$ involves 15 trivalent trees with 4 leaves on $\overline{\mathcal{M}}_{0,5}$. Its cohomology class detects the quartic shadow obstruction $o_4(\mathcal{A})$ (line 7644).

Obstacles. The identification of the formality obstruction tower with the shadow Postnikov tower requires showing that the Harrison cohomology controlling L_∞ formality maps isomorphically to the shadow algebra $\mathcal{A}^{\mathrm{sh}}$. This is plausible but not automatic: the shadow algebra is defined as $H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$ (Definition `def:shadow-algebra`), which is the cyclic deformation complex, not the Harrison complex. The two coincide for commutative algebras but differ in general.

Additionally, the genus contributions complicate the story: $\ell_1^{(g)}$ (the BV operator at genus g) mixes with the genus-0 formality obstructions. A clean statement would need to separate the genus-0 formality from the genus-g corrections.

Utility Assessment. **Very high utility, moderate difficulty.** This is the most satisfying answer to the shadow tower termination question. It gives a dictionary:

Shadow tower	**Formality theory**
κ (arity 2)	Binary Lie bracket
$\mathfrak{C}$ (arity 3)	Ternary Massey product
$\mathfrak{Q}$ (arity 4)	Quaternary formality obstruction
$r_{\max} < \infty$	Finite-stage formality
$r_{\max} = \infty$	Intrinsic non-formality

The Virasoro tower being infinite means: the convolution L_∞-algebra of Vir_c is *intrinsically non-formal.* The Heisenberg tower terminating at arity 2 means: the convolution L_∞-algebra is intrinsically formal (a genuine dg Lie algebra, no higher brackets needed). This is exactly the content of the shadow archetype classification.

6 The Key Question: Operadic Explanation of the Termination Pattern

Let me combine the threads. The shadow tower termination pattern

$$\mathcal{H}@2, \quad \widehat{\mathfrak{g}}@3, \quad \beta\gamma@4, \quad \mathrm{Vir}@\infty$$

admits the following unified operadic explanation, synthesizing Directions C and E:

Proposal 1 (The Operadic Complexity Theorem)**.** For a chiral Koszul algebra $\mathcal{A}$ over $\mathcal{P}^{\mathrm{ch}}$, define:

1. The A_∞-*depth* $d_\infty(\mathcal{A})$ = the largest n such that $m_n^{\mathrm{tr}} \neq 0$ in the minimal A_∞ model of $\mathcal{A}$.
2. The L_∞-*formality level* $f_\infty(\mathcal{A})$ = the largest n such that $\ell_n^{(0),\mathrm{tr}} \neq 0$ in the transferred quantum L_∞ brackets on the convolution algebra.
3. The *shadow depth* $r_{\max}(\mathcal{A})$ = the termination arity of the shadow Postnikov tower.

Then:

$$r_{\max}(\mathcal{A}) = d_\infty(\mathcal{A}) = f_\infty(\mathcal{A}).$$

Verification on examples:

- **Heisenberg**: $d_\infty = 2$ (commutative, all $m_n = 0$ for $n \geq 3$); $f_\infty = 2$ (convolution is a dg Lie algebra, no higher brackets); $r_{\max} = 2$. ✓

- **Affine**: $d_\infty = 3$ (the Lie bracket gives $m_2 \neq 0$; the Koszul resolution is minimal so $m_3 = 0$ on the minimal model, but the cubic shadow $\mathfrak{C} = \kappa(x, [y, z])$ comes from the *convolution* ℓ_3 which is chain-level nontrivial even though cohomologically trivial at genus 0—wait. Actually, $r_{\max} = 3$ means $o_4 = 0$, which is the Jacobi identity. The A_∞-depth is 2 (binary Lie bracket), not 3. The cubic shadow comes from composing m_2 with itself, not from m_3. So we need $d_\infty(\mathcal{A})$ to count the "effective arity" including compositions.)

  Correction: the right notion is not m_n^{tr} alone, but the *effective operadic arity* = the arity at which the graph sums from all vertices of the minimal model can generate nontrivial contributions. For affine: m_2 is nonzero (the Lie bracket), and graphs with two binary vertices sewn together give arity-3 contributions. But graphs with three binary vertices give arity-4 contributions, and $o_4 = 0$ because the Jacobi identity is an operadic relation.

  So the refined statement is: $r_{\max}(\mathcal{A})$ is the arity at which the *operadic relations* of $\mathcal{P}^{\mathrm{ch}}$ kill all further graph-sum contributions. For Lie: the Jacobi identity is the unique relation in arity 3 of the Lie operad, and it kills all arity-4 contributions. This is $r_{\max} = 3$.

- $\beta\gamma$: The A_∞ minimal model has $m_2 = 0$ (no Lie bracket: this is a free field), $m_3 \neq 0$ (from the normal-ordered product and the OPE contact terms). Graphs with one ternary vertex and edges give contributions at arities 3 and 4. The quartic shadow $\mathfrak{Q}$ is nonzero, but $o_5 = 0$ by rank-one abelian rigidity. $r_{\max} = 4$.

- **Virasoro**: $m_2 \neq 0$ (the Virasoro bracket), the composite field $:TT:$ generates effective m_4-type contributions, and the central charge prevents the graph sums from terminating. The obstruction $\{C, Q\}_H \neq 0$ forces $\mathrm{Sh}_5 \neq 0$, and by induction the tower is infinite. $r_{\max} = \infty$.

Conjecture 6 (Refined operadic complexity)**.** $r_{\max}(\mathcal{A})$ *equals the* relational arity *of* $\mathcal{A}$ *as a* $\mathcal{P}^{\mathrm{ch}}$*-algebra: the minimal* r *such that all defining relations of* $\mathcal{A}$ *(including the operadic relations of* $\mathcal{P}^{\mathrm{ch}}$ *and the specific OPE relations of* $\mathcal{A}$*) can be expressed in terms of operations of arity* $\leq r$*, AND the Jacobi-type higher syzygies generated by these relations kill all obstruction classes* $o_{r+1}, o_{r+2}, \ldots$ *simultaneously.*

This is the chiral analogue of the following classical fact: for a quadratic algebra $A = T(V)/(R)$ *with* $R \subset V^{\otimes 2}$*, the Koszul property is equivalent to the quadratic relations* R *generating all syzygies. For a cubic algebra (relations in* $V^{\otimes 3}$*), one needs* $r = 3$*. The "relational arity" is the operadic version of the relation degree.*

7 Synthesis and Recommendations

7.1 Priority ranking

1. **Direction E** (Formality characterization): highest priority. The identification of the shadow tower with the formality obstruction tower for the convolution L_∞-algebra is

the cleanest answer to the termination question and is closest to being provable with existing technology.

2. **Direction C** (Operadic complexity): closely related to Direction E. The graph-sum analysis connecting A_∞-depth to shadow termination should be developed in parallel. This is the computational workhorse.

3. **Direction A** (Operadic inheritance): important for the foundational architecture. Proving that $\mathcal{P}^{\mathrm{ch}}$ is Koszul (using Kontsevich formality) would ground the entire PBW framework in operadic theory.

4. **Direction D** (Genus-dependent Koszul dual): most ambitious, connecting to deep questions in algebraic geometry (formality of configuration spaces on curves). Long-term.

5. **Direction B** (Distributive law): most speculative. The modified version (distributive law on associated graded, shadow tower as lifting obstruction) is viable but not yet precise enough to state as a theorem.

7.2 What the monograph should add

I recommend the following additions:

1. A remark in `chiral_koszul_pairs.tex`, after Theorem `thm:pbw-koszulness-criterion`, stating that the PBW criterion is a *consequence* of the Koszulness of the chiral operad $\mathcal{P}^{\mathrm{ch}}$, in the same way that the E_1-chiral Koszul duality (Theorem `thm:e1-chiral-koszul-duality`) is a consequence of the Koszulness of Ass.

2. A conjecture in `higher_genus_modular_koszul.tex`, after Definition `def:shadow-postnikov-tower`, stating the operadic complexity conjecture (Conjecture **??**): the shadow termination arity equals the A_∞-depth of the minimal model.

3. A remark connecting the shadow Postnikov tower to the formality obstruction tower for the convolution L_∞-algebra (Conjecture **??**). The key formula is already in the monograph (Construction `constr:explicit-convolution-linfty`): the quantum L_∞ brackets $\ell_n^{(g)}$ are the Feynman expansion, and their vanishing on cohomology is formality.

7.3 The bottom line

The shadow tower termination pattern $\mathcal{H}$@2, $\widehat{\mathfrak{g}}$@3, $\beta\gamma$@4, Vir@∞ is not an accident. It measures the *operadic complexity* of the chiral algebra: the depth of its A_∞ minimal model, equivalently, the formality level of its modular convolution L_∞-algebra. Commutative algebras (Heisenberg) are intrinsically formal. Lie algebras (affine) have one layer of non-formality, killed by the Jacobi identity. Contact algebras ($\beta\gamma$) have two layers, killed by abelian rigidity. The Virasoro algebra, with its quartic pole and infinite family of composite fields, is intrinsically non-formal at every level: its modular content is genuinely infinite.

This is the operadic explanation the monograph lacks. The technology to prove it is largely present: the graph-sum formula, the explicit L_∞ brackets, and the obstruction recursion. What remains is the formal argument connecting these ingredients.